Supplementary Materials: Benchmarking Training-Free Token Compression for Large Vision Language Models in Agriculture

Rukshan Karannagoda[1 [0009−0009−5676−1865]], Yadan Luo[1 [0000−0001−6272−2971]], Zijian Wang[1 [0000−0002−7190−9620]], Kaiyu Guo[1 [0000−0002−4187−2839]], Scott C. Chapman[2 [0000−0003−4732−8452]], and Mahsa Baktashmotlagh[1 [0000−0001−5255−8194]]

[1] School of Electrical Engineering and Computer Science, The University of Queensland.
{r.karannagoda, y.luo, zijian.wang, kaiyu.guo, m.baktashmotlagh}@uq.edu.au
[2] School of Agriculture and Food Sustainability, The University of Queensland.
scott.chapman@uq.edu.au

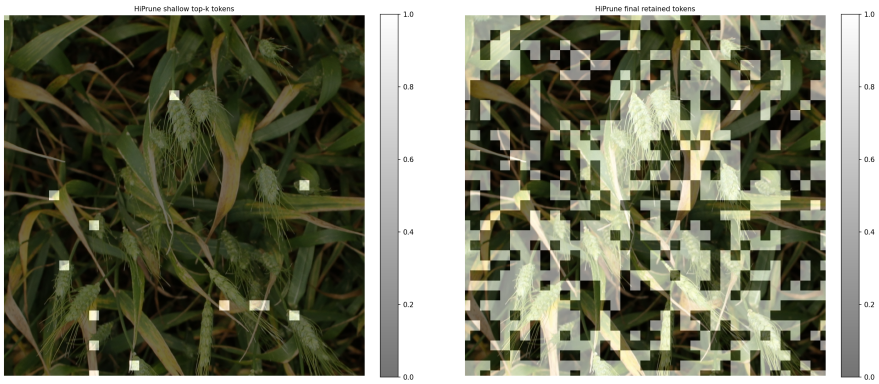
This document provides supplementary material for the manuscript 'Benchmarking Training-Free Token Compression for Large Vision Language Models in Agriculture', submitted to the Australasian Joint Conference on Artificial Intelligence.

Section 1 provides detailed visualizations of the token compression mechanisms for four representative methods, and Section 2 depicts efficiency-performance radar charts and scatter plots. Section 3 examines robustness under fourteen simulated weather and sensor degradation conditions across both the Qwen 2.5-VL 3B and 7B backbones, with and without compression applied. Together these materials substantiate the efficiency analysis and robustness claims made in the main paper.

1 Detailed Visualizations of Token Compression Methods

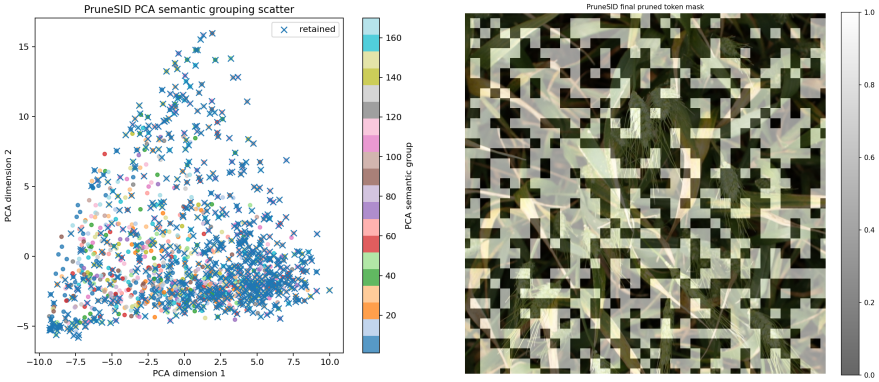
Figures 1–4 visualize the internal token selection mechanism of four methods at 50% retention, pairing each method's intermediate representation (clustering, graph, or saliency map) with its final binary retention mask. These visualizations illustrate why spatially-aware methods such as LLaVA-Scissor and HiPrune preserve fine-grained structure better than semantic-merging approaches: their retained tokens tile the image more uniformly rather than collapsing into high-attention regions.

The subsequent radar plots Figure 5 and scatter plots (Figures 6–9) show the joint efficiency–accuracy as well as memory reduction trade-offs across all ten methods for each task.



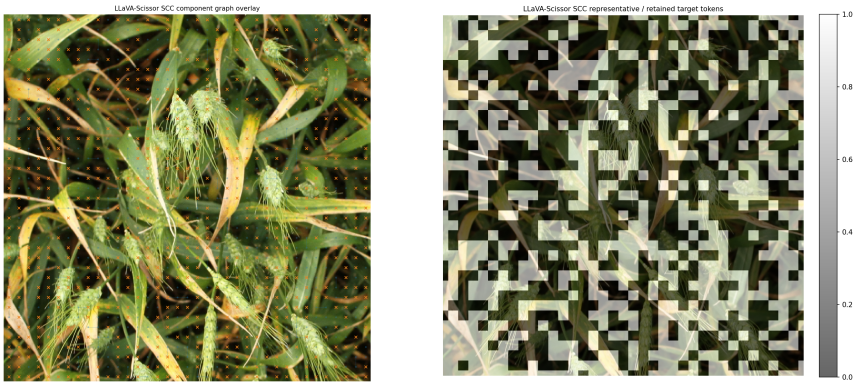
(a) (b)

Fig. 1: Visualization of the HiPrune method, and the corresponding token retention map. 50% of the most semantically diverse tokens are retained.



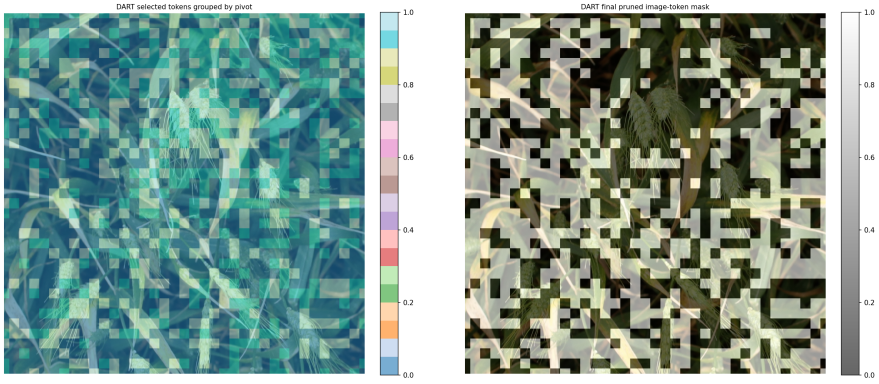
(a) (b)

Fig. 2: Semantic PCA clustering of the PruneSID method, and the corresponding token retention map. 50% of the most semantically diverse tokens are retained.



(a) (b)

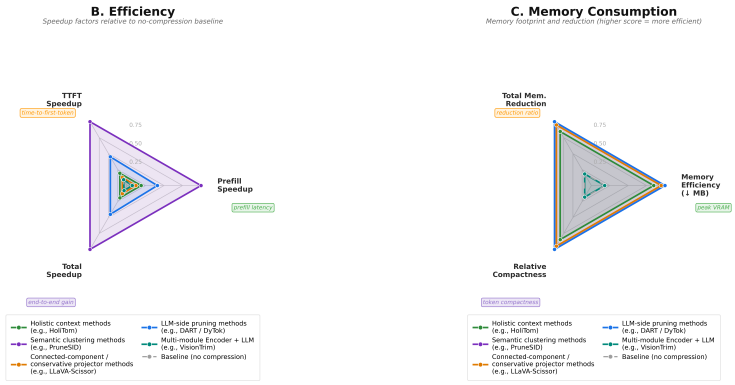
Fig. 3: LLaVA-Scissor spatial similarity graph and token retention map. 50% of the most semantically diverse tokens are retained.



(a) (b)

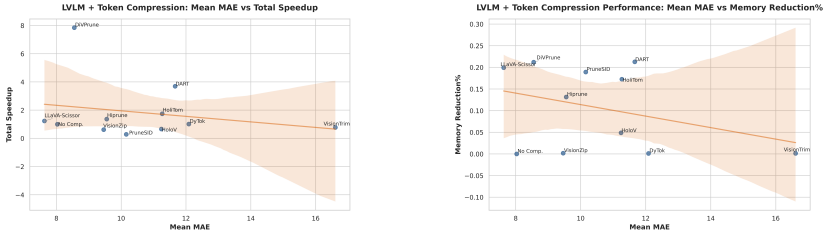
Fig. 4: DART token dissimilarity graph and token retention map. 50% of the most semantically diverse tokens are retained.

2 Training-Free Token Compression Efficiency vs. Accuracy Evaluation



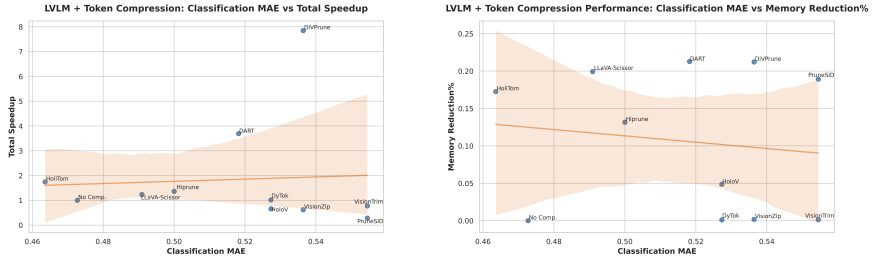
(a) (b)

Fig. 5: Radar chart showing the comparative latency (a) and memory reduction (b) of HoliTom, PruneSID, LLaVA-Scissor, DART, and VisionTrim. Each axis represents an evaluation subset, where a larger enclosed area indicates superior performance on that task.



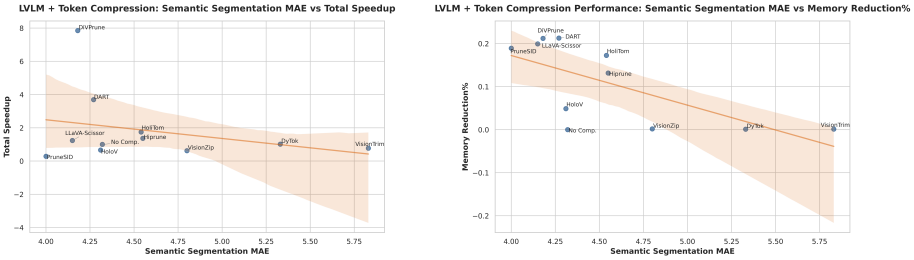
(a) (b)

Fig. 6: Mean Performance (MAE) vs. Speedup comparison (7a) and Memory Reduction% (7b) of fine-tuned Qwen 2.5-VL model with and without various token compression methods.



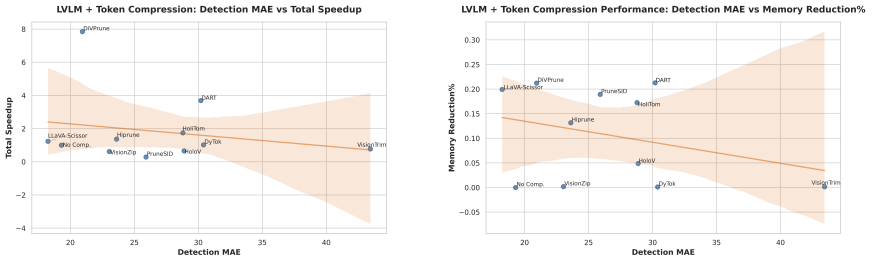
(a) (b)

Fig. 7: Classification Task Performance (MAE) vs. Speedup comparison (7a) and Memory Reduction% (7b) of fine-tuned Qwen 2.5-VL model with and without various token compression methods.



(a) (b)

Fig. 8: Semantic Segmentation Performance (MAE) vs. Speedup comparison (8a) and Memory Reduction% (8b) of fine-tuned Qwen 2.5-VL model with and without various token compression methods.



(a) (b)

Fig. 9: Detection Performance (MAE) vs. Speedup comparison (9a) and Memory Reduction% (9b) of fine-tuned Qwen 2.5-VL model with and without various token compression methods.

Table 1: MAE of fine-tuned Qwen2.5-VL 3B with and without token compression on growth stage classification (Classification), organ percentage estimation (Sem. Segmentation), and spike counting (Detection). Mean Average Error (MAE) is used throughout as the evaluation metric. Best result per column is in **bold**. Results that improve on the baseline are underlined.

Method	Classification	Sem. Segmentation	Detection
Baseline (no compression)	0.4545	5.06	19.02
DART	0.5545	5.54	27.16
DivPrune	**0.4455**	5.00	20.18
DyTok	0.5545	5.54	27.16
HiPrune	0.4909	5.36	23.02
HoliTom	0.5091	5.64	26.93
HoloV	0.5273	**4.31**	28.88
LLaVA-Scissor	0.5091	5.15	20.77
PruneSID	0.5273	5.09	23.82
VisionTrim	0.5545	6.17	31.02
VisionZip	0.4909	5.65	21.82

2.1 Diagnosing Method Failure via Error Decomposition

Table 2 decomposes spike-counting error into RMSE and directional bias (over- vs. under-count) across all methods. Under-counting dominates universally, confirming that no training-free method fully escapes spike merging; what separates stronger from weaker methods is the magnitude of the per-image miss rather than its direction. Figures 10a–10c plot predicted versus ground-truth spike counts for the three weakest performers, DyTok, VisionTrim, and DART.

Table 2: Spike-counting error decomposition: RMSE and % of the 686-image test set over-/under-counted, sorted by ascending RMSE. Undercounting dominates universally; RMSE separates methods by error magnitude, not direction.

Token Compression Method	RMSE	Overcounted (%)	Undercounted (%)
LLaVA-Scissor	**19.65**	0.0	100.0
DivPrune	25.99	3.0	90.5
HiPrune	30.61	0.5	94.0
VisionZip	32.23	3.0	91.0
PruneSID	32.99	0.5	94.0
HoloV	35.31	0.5	94.0
HoliTom	35.83	0.5	94.0
DART	37.22	0.5	94.0
DyTok	38.12	1.0	93.5
VisionTrim	**51.59**	0.0	94.0

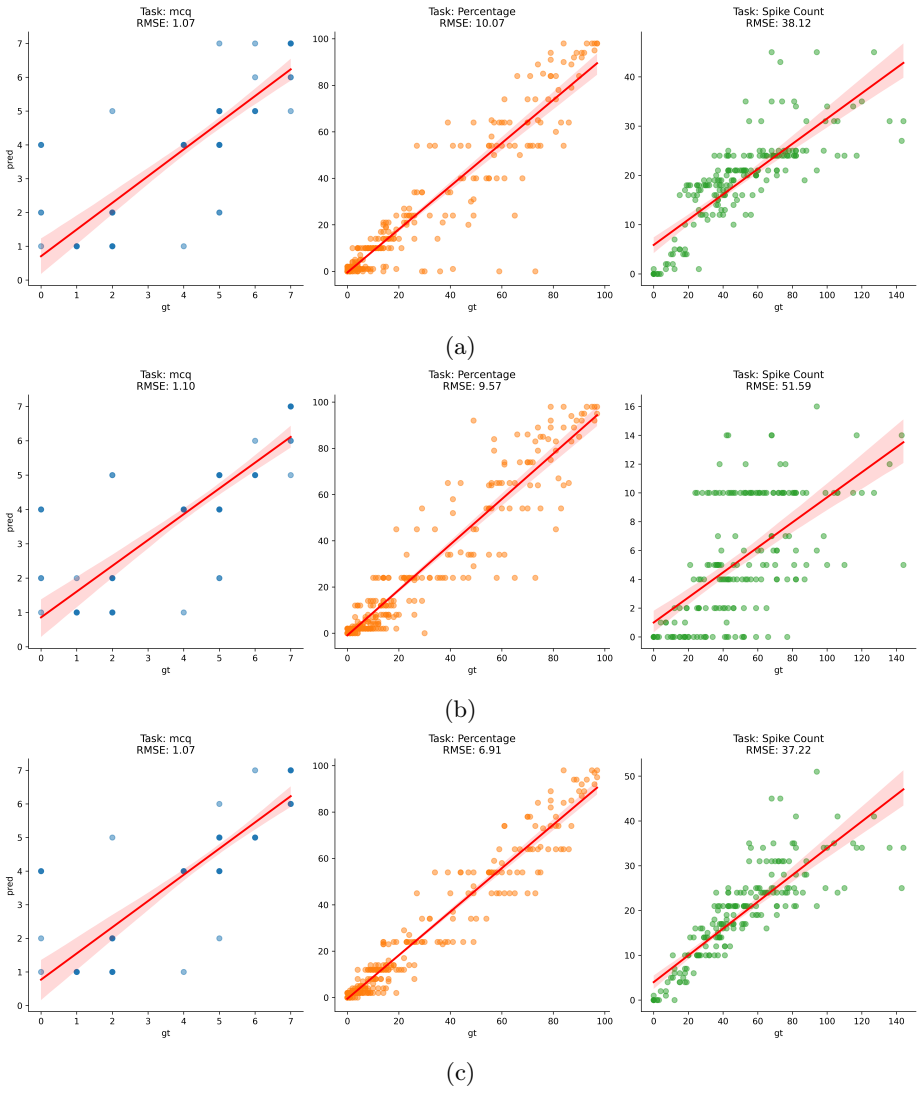


Fig. 10: Regression plots of predictions of the weakest models: DyTok (a), VisionTrim (b), and DART (c). Detection tasks incur the highest per-image error, and thus the highest RMSE.

3 Robustness to different weather and sensor conditions

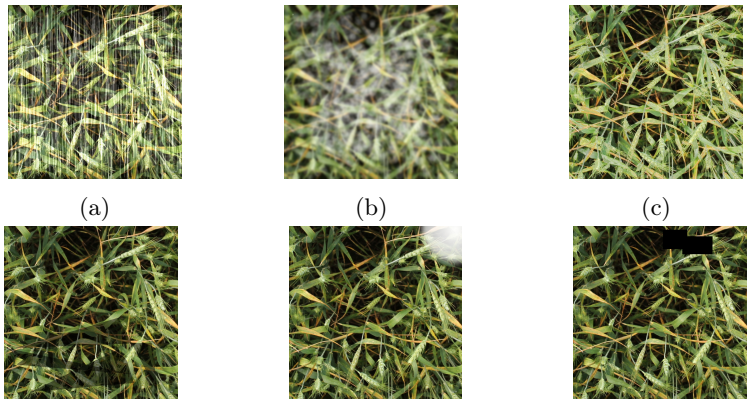


Fig. 11: Augmented views tested against Qwen 2.5-VL 3B and 7B and 6 token compression methods: (a) Torrential Rain, (b) Thick Fog, (c) Snow, (d) Shadow, (e) Sun flare, and (f) Dropped pixels.

Figure 11 illustrates the fourteen augmentation conditions applied at inference time to simulate UAV field degradation: two rain intensities (drizzle, torrent), three fog densities (fog, fog×0.5, fog×5) , two shadow dimensions, snow, sun flare (alpha blending, physics-based blending), and dropped pixels from sensor noise. All conditions are evaluated on the same 686-sample test split used in the main paper; no model re-training is performed. The tables below report MAE separately for classification, semantic segmentation, and detection, which have very different absolute error scales and respond differently to visual degradation: detection (spike counting) is by far the most sensitive, as blurring and occlusion directly collapse the spatial separability of dense wheat-head clusters. The original and non-augmented baseline results of the Qwen2.5-VL model may vary by 2-7% due to the preprocessing differences between the non-augmented and augmented inference pipelines.

3.1 Qwen2.5-VL 3B

The 3B model is the more practically deployable backbone given its lower memory footprint. These results show whether token compression compounds or partially absorbs the degradation introduced by augmentation. A method that performs similarly on original and augmented images offers more reliable field deployment than one whose MAE gap widens under adverse conditions.

Table 3: Effect of Weather and Sensor Augmentation on non-token-compressed Qwen 2.5-VL 3B under Growth Stage MCQ (Classification), Organ Percentage Estimation (Semantic Segmentation), and Spike Counting (Detection) tasks, evaluated with Mean Absolute Error (MAE). Original= non-augmented. d4=4 vertices

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.4727	5.6515	24.73
dropped_pixels	0.5182	5.5848	22.125
fog	0.6273	5.8242	41.365
fog×0.5	0.6364	5.5758	40.85
fog×5	0.7545	10.103	38.02
original	0.4636	4.9576	20.39
rain	0.8818	7.4121	31.855
shadow	0.5	5.7182	23.955
shadow_d4	0.5	5.803	24.12
snow	0.5455	5.0848	19.57
snow×1.2	0.4455	5.0576	19.14
sunflare	0.5636	5.0636	25.035
sunflare_physics	0.5364	5.0303	22.525
torrent	1.0636	9.0667	33.885

Table 4: Effect of Weather and Sensor Augmentation on Qwen 2.5-VL 3B + VisionZip token compression under Growth Stage MCQ (Classification), Organ Percentage Estimation (Semantic Segmentation), and Spike Counting (Detection) tasks, evaluated with Mean Absolute Error (MAE). Original= non-augmented

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.4909	6.0182	24.6000
fog	0.5091	6.4697	41.4550
fog×0.5	0.6091	6.4152	40.7300
fog×5	0.8818	10.3848	37.3950
original	0.4818	5.7818	21.3700
rain	1.0000	7.3030	31.0850
shadow	0.6000	6.7303	24.8250
shadow_d4	0.5727	6.3515	24.9400
snow	0.5273	5.6212	21.3900
snow×1.2	0.5273	5.1667	20.5750
sunflare	0.6364	5.4242	25.2300
sunflare_physics	0.5455	5.6667	24.1200
torrent	1.0455	9.2788	33.9550
dropped_pixels	0.6091	5.9515	23.6250

Table 5: Effect of Weather and Sensor Augmentation on Qwen 2.5-VL 3B + PruneSID token compression under Growth Stage MCQ (Classification), Organ Percentage Estimation (Semantic Segmentation), and Spike Counting (Detection) tasks, evaluated with Mean Absolute Error (MAE). Original= non-augmented

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.5727	5.7091	26.805
dropped_pixels	0.5818	5.6394	25.12
fog	0.6273	5.6788	41.81
fog×0.5	0.6455	5.8818	40.955
fog×5	0.8455	9.0212	38.665
original	0.4909	5.2364	23.325
rain	1	7.0636	31.925
shadow	0.5545	6.0212	26.665
shadow_d4	0.4455	5.8	26.44
snow	0.5545	5.2455	22.895
snow×1.2	0.4909	5.1576	22.63
sunflare	0.6727	5.0727	27.295
sunflare_physics	0.5273	5.2879	25.83
torrent	1.1818	9.4121	34.33

Table 6: Effect of Weather and Sensor Augmentation on Qwen 2.5-VL 3B + LLaVA-Scissor token compression under Growth Stage MCQ (Classification), Organ Percentage Estimation (Semantic Segmentation), and Spike Counting (Detection) tasks, evaluated with Mean Absolute Error (MAE). Original= non-augmented

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.5091	5.8212	24.9600
dropped_pixels	0.5727	5.8788	22.805
fog	0.6636	5.7939	41.5700
fog×0.5	0.6909	5.9152	41.2550
fog×5	0.8182	9.9879	37.9500
original	0.5182	5.3455	20.1400
rain	0.9636	7.3879	31.9000
shadow	0.5636	6.3273	24.5450
shadow_d4	0.5000	6.1000	24.8150
snow	0.5364	5.5727	20.0150
snow×1.2	0.4909	5.3939	19.8000
sunflare	0.6636	5.2545	25.9350
sunflare_physics	0.4818	5.1212	23.2100
torrent	1.1000	9.0061	35.1800

Table 7: Effect of Weather and Sensor Augmentation on Qwen 2.5-VL 3B + HoloV token compression under Growth Stage MCQ (Classification), Organ Percentage Estimation (Semantic Segmentation), and Spike Counting (Detection) tasks, evaluated with Mean Absolute Error (MAE). Original= non-augmented

Augmentation Method	**Classification**	**Semantic Segmentation**	**Detection**
drizzle	0.5545	6.0182	29.990
dropped_pixels	0.5636	5.4000	28.100
fog	0.7455	6.0818	43.720
fog×0.5	0.7273	6.0667	42.800
fog×5	0.9182	10.3939	40.245
original	0.4455	5.0848	27.270
rain	0.9545	7.8485	35.385
shadow	0.5545	5.9515	29.675
shadow_d4	0.5909	5.8939	29.970
snow	0.5636	5.2303	26.015
snow×1.2	0.4364	5.0818	26.715
sunflare	0.6818	5.5424	30.035
sunflare_physics	0.5727	5.1667	29.375
torrent	1.1455	10.0000	36.570

Table 8: Effect of Weather and Sensor Augmentation on Qwen 2.5-VL 3B + DART token compression under Growth Stage MCQ (Classification), Organ Percentage Estimation (Semantic Segmentation), and Spike Counting (Detection) tasks, evaluated with Mean Absolute Error (MAE). Original= non-augmented

Augmentation Method	**Classification**	**Semantic Segmentation**	**Detection**
drizzle	0.5818	5.7848	28.835
dropped_pixels	0.4818	5.6879	26.185
fog	0.6455	5.6818	42.235
fog×0.5	0.6909	5.8424	41.255
fog×5	0.9091	9.4242	38.62
original	0.5636	5.5515	26.66
rain	0.9364	6.9273	33.765
shadow	0.5636	6.1485	27.625
shadow_d4	0.5545	5.9697	27.225
snow	0.5727	5.4091	25.25
snow×1.2	0.4909	5.5212	25.66
sunflare	0.5818	5.6515	28.305
sunflare_physics	0.5545	5.3182	28.09
torrent	1.0091	8.7697	35.36

3.2 Qwen2.5-VL 7B

The equivalent results for the Qwen 2.5-VL 7B backbone are provided below. Comparing against the 3B tables above isolates the effect of model scale on weather and sensor robustness independently of the token compression method applied.

Table 9: Qwen 2.5-VL 7B uncompressed baseline performance (MAE) against different weather and sensor condictions, divided by the three tasks of classification, semantic segmentation, and detection

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.4818	4.7212	23.585
dropped_pixels	0.4545	5.3424	22.1450
fog	0.4364	4.9364	43.85
fog×0.5	0.4455	10.4091	43.525
fog×5	0.7455	4.3606	45.205
original	0.4636	6.3636	19.12
rain	0.6636	4.4061	30.405
shadow	0.4364	4.397	21.215
shadow_d4	0.4727	4.8182	21.655
snow	0.4727	4.5121	21.725
snow×1.2	0.4545	4.6909	17.7
sunflare	0.5	4.3667	22.785
sunflare_physics	0.4818	8.1303	20.325
torrent	0.6909	8.7697	34.395

Table 10: Qwen 2.5-VL 7B + LLaVA-Scissor performance (MAE) against different weather and sensor condictions, divided by the three tasks of classification, semantic segmentation, and detection

Augmentation Method	Classsification	Semantic Segmentation	Detection
drizzle	0.5091	4.7303	23.4150
fog	0.5091	5.2848	43.9750
fog×0.5	0.4364	4.7152	43.8100
fog×5	0.7364	10.7788	45.9550
original	0.5182	4.0242	17.7900
rain	0.6636	5.6000	31.1400
shadow	0.5182	4.2273	20.3650
shadow_d4	0.5000	4.1545	21.1000
snow	0.5091	4.6636	23.0600
snow×1.2	0.5091	4.2606	17.2450
sunflare	0.5364	4.3303	22.6250
sunflare_physics	0.4818	4.2606	19.2100
torrent	0.7636	7.7848	35.5400

Table 11: Qwen 2.5-VL 7B + VisionZip performance (MAE) against different weather and sensor condictions, divided by the three tasks of classification, semantic segmentation, and detection

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.5000	5.1576	26.2750
fog	0.5091	6.0848	44.2200
fog×0.5	0.4091	5.1091	43.8700
fog×5	0.8455	10.4879	45.9100
original	0.5182	4.7818	22.9450
rain	0.6818	6.2727	32.6200
shadow	0.4727	4.9182	24.6250
shadow_d4	0.4273	4.8121	24.7400
snow	0.5636	4.7970	25.9750
snow×1.2	0.4909	4.9697	22.1300
sunflare	0.5091	4.8091	26.3850
sunflare_physics	0.5364	4.8848	24.0300
torrent	0.8000	7.6788	36.9900

Table 12: Qwen 2.5-VL 7B + HoloV performance (MAE) against different weather and sensor condictions, divided by the three tasks of classification, semantic segmentation, and detection

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.5000	4.6333	31.98
fog	0.4636	5.3909	45.07
fog×0.5	0.4455	4.7909	45.05
fog×5	0.7818	11.0061	46.435
original	0.5273	4.3061	28.88
rain	0.6636	6.0545	36.54
shadow	0.5182	4.6515	29.785
shadow_d4	0.4727	4.5242	29.71
snow	0.5636	5.0303	31.29
snow×1.2	0.4909	4.6364	28.615
sunflare	0.5545	4.5121	31.435
sunflare_physics	0.5364	4.4273	29.815
torrent	0.7364	8.0152	39.52

Table 13: Qwen 2.5-VL 7B + PruneSID performance (MAE) against different weather and sensor condictions, divided by the three tasks of classification, semantic segmentation, and detection

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.4909	4.8000	28.9550
dropped_pixels	0.4727	4.0121	25.7900
fog	0.5364	5.4455	43.7600
fog×0.5	0.4727	4.8364	43.7050
fog×5	0.7273	10.3303	45.6350
original	0.5455	3.9212	25.5550
rain	0.6818	5.9212	34.7450
shadow	0.4727	4.0364	26.2250
shadow_d4	0.5000	4.2970	26.8050
snow	0.5818	4.3000	28.7100
snow×1.2	0.5273	4.1061	24.9450
sunflare	0.5364	3.9667	28.1100
sunflare_physics	0.5273	4.1182	26.1350
torrent	0.6909	7.2424	37.9300

Table 14: Qwen 2.5-VL 7B + DART performance (MAE) against different weather and sensor condictions, divided by the three tasks of classification, semantic segmentation, and detection

Augmentation Method	Classification	Semantic Segmentation	Detection
drizzle	0.5091	4.9545	32.4900
dropped_pixels	0.4818	4.1364	30.0100
fog	0.4727	5.8576	44.8400
fog×0.5	0.5000	5.2515	44.5100
fog×5	0.8273	10.7606	46.2750
original	0.5182	4.1636	29.8350
rain	0.6636	6.1061	36.4150
shadow	0.5364	4.4485	30.2450
shadow_d4	0.4636	4.2515	30.3300
snow	0.5182	4.8212	32.6400
snow×1.2	0.5545	4.1152	29.5250
sunflare	0.5545	4.5606	31.4950
sunflare_physics	0.4909	4.2333	30.7700
torrent	0.6909	8.0879	38.7200

The 7B results serve as the upper-bound reference for model capacity. Comparing the 7B and 3B tables isolates the contribution of model scale to robustness independently of compression. In general, the 7B baseline is more resilient to moderate augmentations (drizzle, shadow) but similarly brittle under the most severe conditions (torrent, thick fog) regardless of compression method applied.